

SPECIAL SUMMER INTEGRATED LEARNING PROGRAMME MATHEMATICS

Functions

Week 1 Day 2 PM

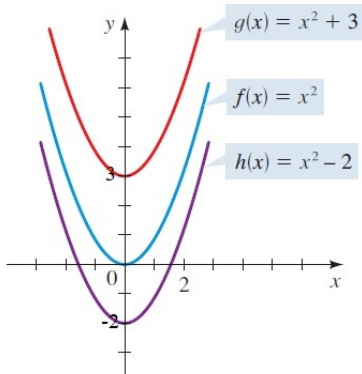


SINGAPORE UNIVERSITY OF
TECHNOLOGY AND DESIGN

OUTLINE

- #### ④ Summary

VERTICAL SHIFTING - EXAMPLE



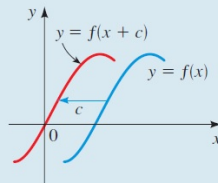
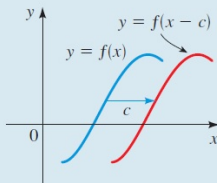
HORIZONTAL SHIFTING

HORIZONTAL SHIFTS OF GRAPHS

Suppose $c > 0$.

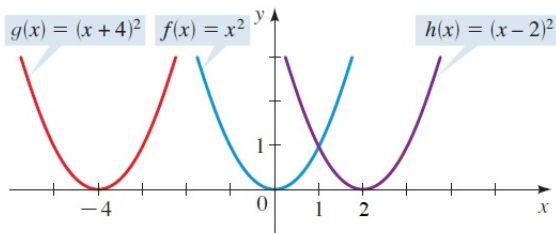
To graph $y = f(x - c)$, shift the graph of $y = f(x)$ to the right c units.

To graph $y = f(x + c)$, shift the graph of $y = f(x)$ to the left c units.



The shifting is **opposite** to the vertical case because c is added/subtracted on the same side as x . If we can make x as the subject, then the original function is $y = f(x) \Leftrightarrow x = f^{-1}(y)$ (we'll learn f^{-1} in a later **section** today) and the new function is $y = f(x + c) \Leftrightarrow x + c = f^{-1}(y) \Leftrightarrow x = f^{-1}(y) - c$. So $+c$ on the same side as x is actually subtracting c from the old x -value.

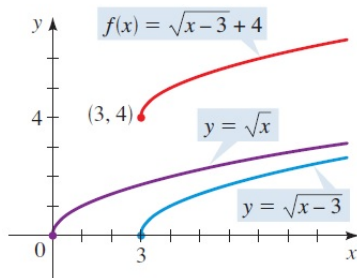
HORIZONTAL SHIFTING - EXAMPLE



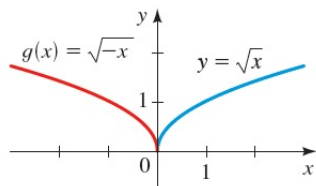
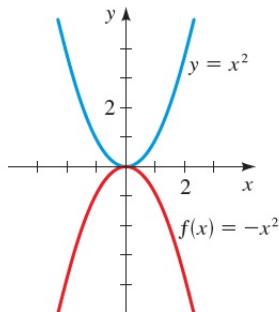
COMBINING VERTICAL AND HORIZONTAL SHIFTING

When we want to sketch the graph of $y = f(x + c) + d$ using the graph of $y = f(x)$, we can think of $g(x) := f(x + c)$. So we first do a horizontal shift to get the graph of $y = f(x + c) = g(x)$, after which we do a vertical shift to get $y = g(x) + d = f(x + c) + d$. You can also do the vertical shift first followed by the horizontal shift.

E.g.:



REFLECTING GRAPHS - EXAMPLE



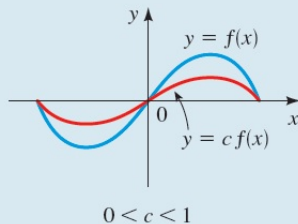
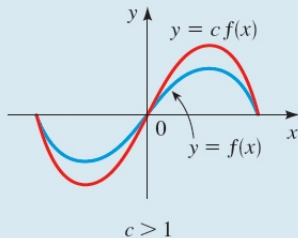
VERTICAL STRETCHING AND SHRINKING

VERTICAL STRETCHING AND SHRINKING OF GRAPHS

To graph $y = cf(x)$:

If $c > 1$, stretch the graph of $y = f(x)$ vertically by a factor of c .

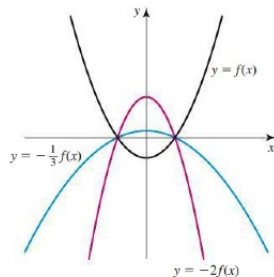
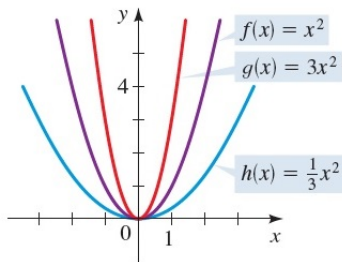
If $0 < c < 1$, shrink the graph of $y = f(x)$ vertically by a factor of c .



Remark: Note that the x -intercepts will not change. That's because stretching and shrinking $y = 0$ is still $y = 0$.

VERTICAL STRETCHING AND SHRINKING - EXAMPLE

For $y = cf(x)$ where $c < 0$, we can interpret it as $y = -(|c|f(x))$. So we do the usual vertical stretching and shrinking for the scalar $|c|$, after which we do a vertical reflection for the $-$ sign.



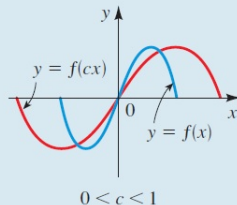
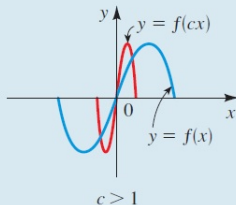
HORIZONTAL STRETCHING AND SHRINKING

HORIZONTAL SHRINKING AND STRETCHING OF GRAPHS

To graph $y = f(cx)$:

If $c > 1$, shrink the graph of $y = f(x)$ horizontally by a factor of $1/c$.

If $0 < c < 1$, stretch the graph of $y = f(x)$ horizontally by a factor of $1/c$.

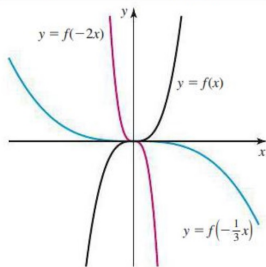
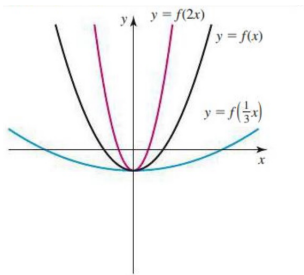


Remarks:

- Same reasoning as before, it's **opposite** to the vertical case because the scalar c is on the same side as x .
- Note that the y -intercepts will not change. That's because stretching and shrinking $x = 0$ is still $x = 0$.

HORIZONTAL STRETCHING AND SHRINKING - EXAMPLE

For $y = f(cx)$ where $c < 0$, we can interpret it as $y = f(-(|c|x))$. So we do the usual horizontal stretching and shrinking for the scalar $|c|$, after which we do a horizontal reflection for the $-$ sign.



Try out this [Geogebra applet](#)!

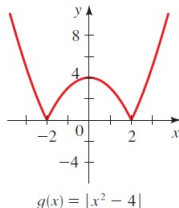
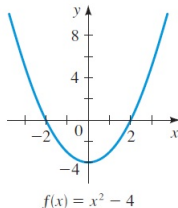
ABSOLUTE VALUE

GRAPHING $y = |f(x)|$

To graph $y = |f(x)|$, **keep** the part of the graph of $y = f(x)$ that is **above** the x -axis, and **vertically reflect** the part of the graph of $y = f(x)$ that is **below** the x -axis about the x -axis.

This is because when the graph of $y = f(x)$ is above the x -axis, it means $y = f(x) > 0$ and so $y = |f(x)| = f(x)$. When the graph of $y = f(x)$ is below the x -axis, it means $y = f(x) < 0$ and so $y = |f(x)| = -f(x)$, which means we need to do a vertical reflection about the x -axis.

E.g.:



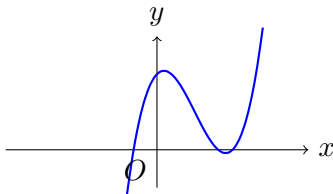
ABSOLUTE VALUE

GRAPHING $y = f(|x|)$

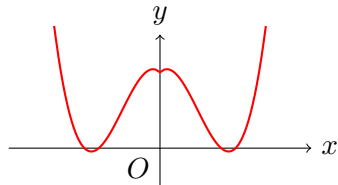
To graph $y = f(|x|)$, **keep** the part of the graph of $y = f(x)$ for $x \geq 0$, and **horizontally reflect this part** about the y -axis.

This is because when $x \geq 0$, $|x| = x$, so $y = f(|x|) = f(x)$. When $x < 0$, $|x| = -x$ and so $y = f(|x|) = f(-x)$, which means we need to do a horizontal reflection about the y -axis.

E.g.:



$$y = x^3 - 3x^2 + x + 2$$



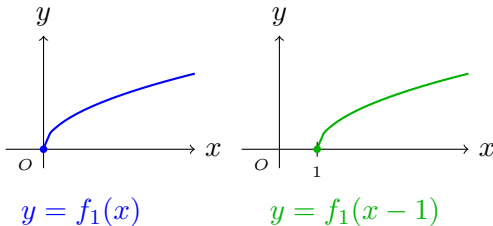
$$y = |x|^3 - 3|x|^2 + |x| + 4$$

COMBINATION

In applications, we will often encounter functions that are obtained from a combination of transformations. Instead of deriving a complicated recipe for graphing such functions, we resort to simply breaking the functions into a series of transformations and perform the series of corresponding changes in the graph. We illustrate this with an example.

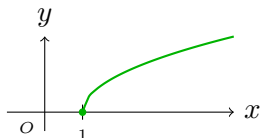
E.g.: Suppose we want to graph $y = 1 - 2\sqrt{|3x| - 1}$, we'll begin with the familiar function $y = f_1(x) = \sqrt{x}$.

1. $y = f_1(x - 1) = \sqrt{x - 1} =: f_2(x)$: horizontal shifting to the right by 1.

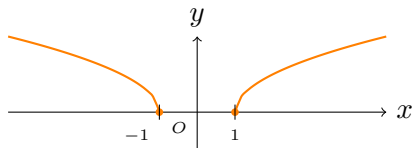


COMBINATION

2. $y = f_2(|x|) = \sqrt{|x| - 1} =: f_3(x)$: reflect the $x \geq 0$ part about the y -axis.

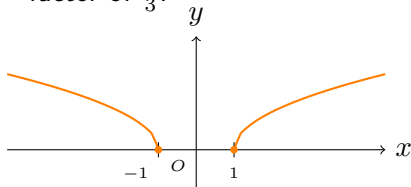


$$y = f_2(x)$$

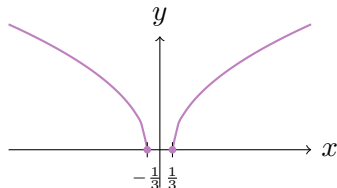


$$y = f_2(|x|)$$

3. $y = f_3(3x) = \sqrt{|3x| - 1} =: f_4(x)$: horizontal shrinking by a factor of $\frac{1}{3}$.



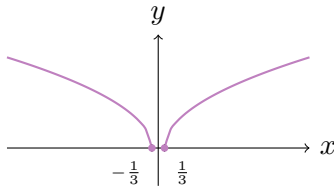
$$y = f_3(x)$$



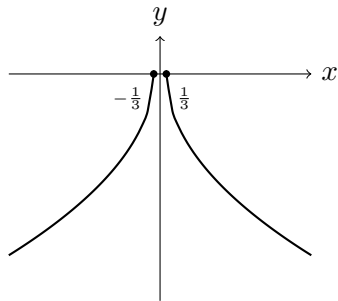
$$y = f_3(3x)$$

COMBINATION

4. $y = -2f_4(x) = -2\sqrt{|3x| - 1} =: f_5(x)$: vertical reflection and stretching by a factor of 2.



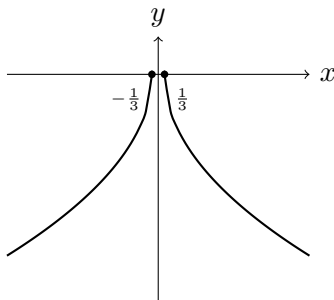
$$y = f_4(x)$$



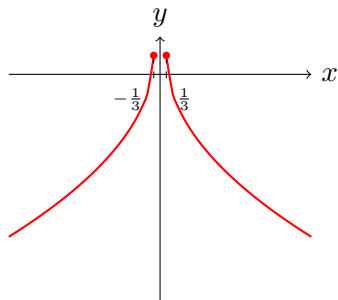
$$y = -2f_4(x)$$

COMBINATION

5. $y = f_5(x) + 1 = 1 - 2\sqrt{|3x| - 1}$: vertical shifting up by 1.



$$y = f_5(x)$$



$$y = f_5(x) + 1$$

ACTIVITY 1

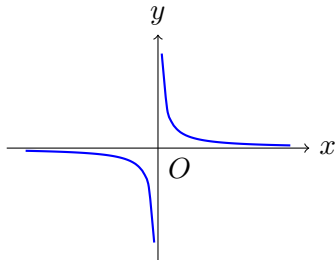
By applying the properties of transformations of functions, sketch the graphs of the following functions.

(a) $y = 2(x - 7)^2 + 1$

(b) $y = \sqrt{3 - x} + 4$

(c) $y = \frac{x+2}{x+3} = \frac{(x+3)-1}{x+3} = 1 - \frac{1}{x+3}$.

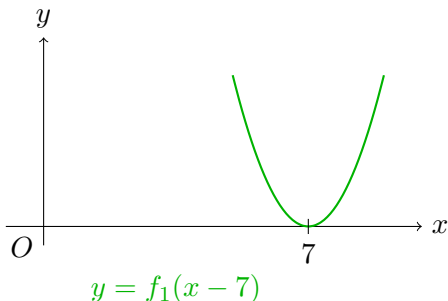
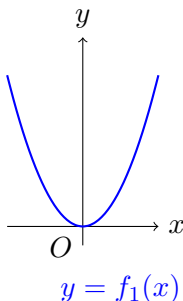
Hint: the graph of $y = \frac{1}{x}$ is as follows:



ACTIVITY 1: SOLUTION

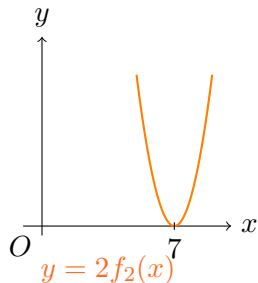
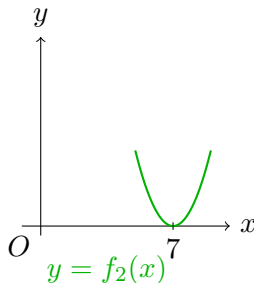
(a) To graph $y = 2(x - 7)^2 + 1$, we'll begin with the familiar function $y = f_1(x) = x^2$.

1. $y = f_1(x - 7) = (x - 7)^2 =: f_2(x)$: horizontal shifting to the right by 7.



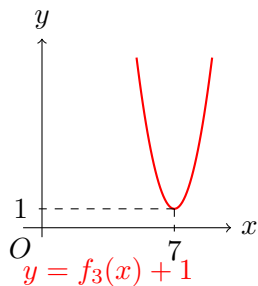
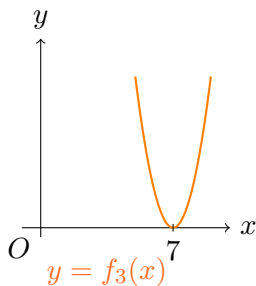
ACTIVITY 1: SOLUTION

- (a) 2. $y = 2f_2(x) = 2(x - 7)^2 =: f_3(x)$: vertical stretching by a factor of 2.



ACTIVITY 1: SOLUTION

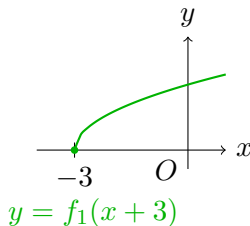
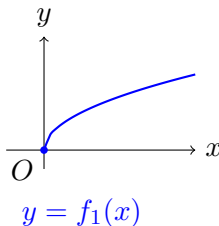
- (a) 3. $y = f_3(x) + 1 = 2(x - 7)^2 + 1$: vertical shifting up by 1.



ACTIVITY 1: SOLUTION

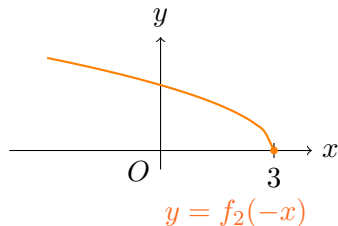
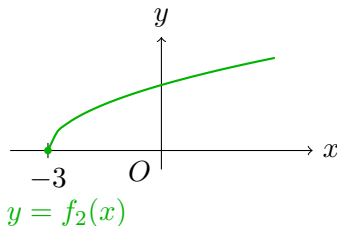
(b) To graph $y = \sqrt{3-x} + 4$, we'll begin with the familiar function $y = f_1(x) = \sqrt{x}$.

1. $y = f_1(x+3) = \sqrt{x+3} =: f_2(x)$: horizontal shifting to the left by 3.



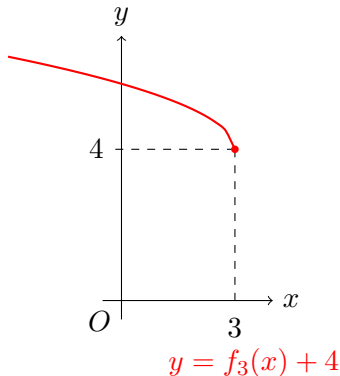
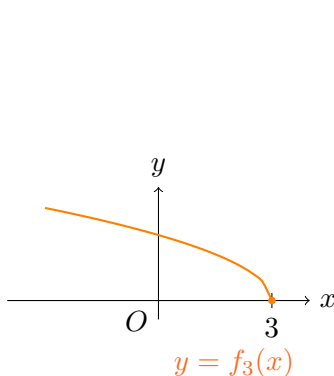
ACTIVITY 1: SOLUTION

- (b) 2. $y = f_2(-x) = \sqrt{-x+3} =: f_3(x)$: vertical reflection about the y -axis.



ACTIVITY 1: SOLUTION

- (b) 3. $y = f_3(x) + 4 = \sqrt{-x + 3} + 4$: vertical shifting up by 4.

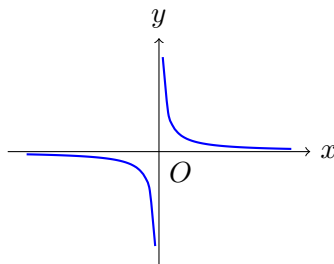


ACTIVITY 1: SOLUTION

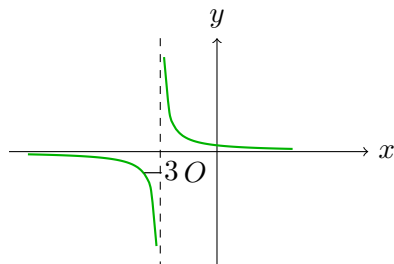
(c) To graph $y = 1 - \frac{1}{x+3}$, we'll begin with the familiar function

$$y = f_1(x) = \frac{1}{x}.$$

1. $y = f_1(x+3) = \frac{1}{x+3} =: f_2(x)$: horizontal shifting to the left by 3.



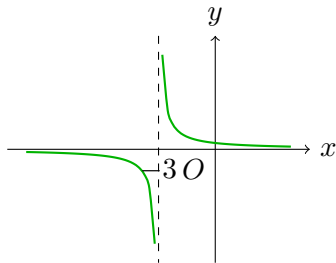
$$y = f_1(x)$$



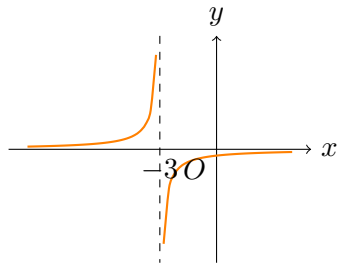
$$y = f_1(x+3)$$

ACTIVITY 1: SOLUTION

- (c) 2. $y = -f_2(x) = -\frac{1}{x+3} =: f_3(x)$: vertical reflection about the x -axis.



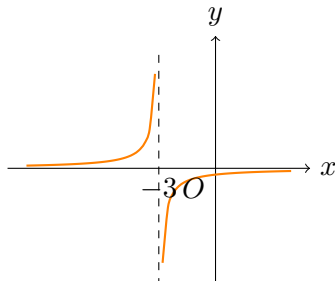
$$y = f_2(x)$$



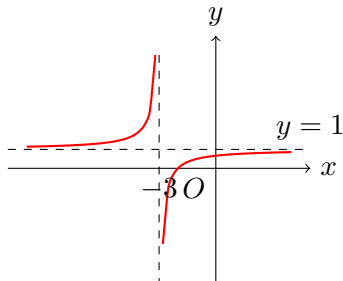
$$y = -f_2(x)$$

ACTIVITY 1: SOLUTION

- (c) 3. $y = f_3(x) + 1 = 1 - \frac{1}{x+3}$: vertical shifting up by 1.



$$y = f_3(x)$$



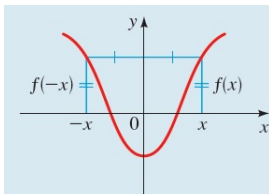
$$y = f_3(x) + 1$$

ODD AND EVEN FUNCTIONS

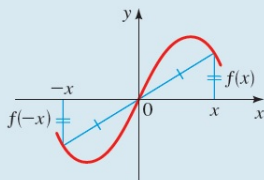
DEFINITION:

Let $f : D \rightarrow \mathbb{R}$ be a function.

- f is **even** if $f(-x) = f(x)$ for all $x \in D$. Geometrically, f is even if its graph is symmetric w.r.t. the y -axis.
- f is **odd** if $f(-x) = -f(x)$ for all $x \in D$. Geometrically, f is odd if its graph is symmetric about the origin.



The graph of an even function is symmetric with respect to the y -axis.



The graph of an odd function is symmetric with respect to the origin.

⚠ There are functions that are neither odd nor even. There is a unique function that is both odd and even.

ACTIVITY 2

Which of the following functions are odd, even, both or neither?

(a) $f(x) = x$ for $x \in \mathbb{R}$.

(b) $g(x) = x + 1$ for $x \in \mathbb{R}$.

(c) $h(x) = 3$ for $x \in \mathbb{R}$.

(d) $j(x) = 0$ for $x \in \mathbb{R}$.

(e) $k(x) = \frac{1}{x}$ for $x \in \mathbb{R} \setminus \{0\}$.

(f) $p(x) = x^n$ for $x \in \mathbb{R}$, where n is a positive odd integer.

(g) $q(x) = x^m$ for $x \in \mathbb{R}$, where m is a positive even integer.

ACTIVITY 2: SOLUTION

- (a) f is odd.
- (b) g is neither.
- (c) h is even.
- (d) j is both odd and even.
- (e) k is odd.
- (f) p is odd.
- (g) q is even.

You can check them by definition. For example:

$$f(-x) = -x = -f(x),$$

so f is odd. But $f(-x) = -x \neq f(x)$, so f is not even. On the other hand,

$$g(-x) = -x + 1,$$

which is not equal to $g(x) = x + 1$ nor $-g(x) = -x - 1$, so g is neither odd nor even.

OUTLINE

- 1 Transformations of Functions
- 2 Combining Functions
- 3 One-to-One, Onto and Inverses
- 4 Summary

SUMS, DIFFERENCES, PRODUCTS, AND QUOTIENTS

Two functions f and g can be combined to form new functions $f + g$, $f - g$, fg , and $\frac{f}{g}$ in a manner similar to the way we add, subtract, multiply, and divide real numbers.

ALGEBRA OF FUNCTIONS

Let f and g be functions with domains A and B . Then the functions $f + g$, $f - g$, fg , and f/g are defined as follows.

$$(f + g)(x) = f(x) + g(x) \quad \text{Domain } A \cap B$$

$$(f - g)(x) = f(x) - g(x) \quad \text{Domain } A \cap B$$

$$(fg)(x) = f(x)g(x) \quad \text{Domain } A \cap B$$

$$\left(\frac{f}{g}\right)(x) = \frac{f(x)}{g(x)} \quad \text{Domain } \{x \in A \cap B \mid g(x) \neq 0\}$$

SUMS, DIFFERENCES, PRODUCTS, AND QUOTIENTS - EXAMPLE

E.g.: Let $f(x) = \frac{1}{x-2}$, $g(x) = \sqrt{x}$. By convention, the domain of f and g are $\mathbb{R} \setminus \{2\}$ and $[0, \infty)$ respectively. So

$$(f+g)(x) = \frac{1}{x-2} + \sqrt{x}, \text{ domain} = \mathbb{R} \setminus \{2\} \cap [0, \infty) = [0, 2) \cup (2, \infty),$$

$$(f-g)(x) = \frac{1}{x-2} - \sqrt{x}, \text{ domain} = \mathbb{R} \setminus \{2\} \cap [0, \infty) = [0, 2) \cup (2, \infty),$$

$$(fg)(x) = \frac{\sqrt{x}}{x-2}, \text{ domain} = \mathbb{R} \setminus \{2\} \cap [0, \infty) = [0, 2) \cup (2, \infty),$$

$$\left(\frac{f}{g}\right)(x) = \frac{1}{\sqrt{x}(x-2)}, \text{ domain} = \mathbb{R} \setminus \{2\} \cap [0, \infty) \setminus \{0\} = (0, 2) \cup (2, \infty).$$

COMPOSITE FUNCTIONS

Let $f(x) = x + 1$, $g(x) = x^2$ be two functions. If we first apply f to x , we get $x + 1$, then we apply g to $x + 1$, we'll get $(x + 1)^2$. This is the idea of composite functions.

$$x \xrightarrow{f} f(x) = x + 1 \xrightarrow{g} g(f(x)) = (x + 1)^2.$$

This is possible if the value $f(x)$ can be input into g , i.e. the value $f(x)$ is in the domain of g .

DEFINITION:

Given two functions $f : A \longrightarrow B$ and $g : C \longrightarrow D$, then the **composite function** $g \circ f$ is defined to be the function

$$\begin{aligned} g \circ f : E &\longrightarrow D \\ x &\longmapsto g(f(x)), \end{aligned}$$

where $E := \{x \in A \mid f(x) \in C\}$.

Notation: $(h \circ g \circ f)(x)$ means $(h \circ (g \circ f))(x) = h(g(f(x)))$.

COMPOSITE FUNCTIONS - EXAMPLES

E.g.: Let $f(x) = 3x + 1$, $g(x) = x^2$. Then the domains of f, g are $\mathbb{R}$

$$(g \circ f)(x) = g(f(x)) = f(x)^2 = (3x + 1)^2,$$

$$\left(\text{ or } (g \circ f)(x) = g(f(x)) = g(3x + 1) = (3x + 1)^2 \right)$$

with domain = $\mathbb{R}$.

$$(f \circ g)(x) = f(g(x)) = 3g(x) + 1 = 3x^2 + 1.$$

$$\left(\text{ or } (f \circ g)(x) = f(g(x)) = f(x^2) = 3x^2 + 1. \right)$$

with domain = $\mathbb{R}$.

⚠ Pay attention to the order of f and g in $g \circ f$ and $f \circ g$. **The one that is closest to x is applied first.**

COMPOSITE FUNCTIONS

Fun facts:

- Fascinating patterns might emerge if we repeatedly compose a function with itself: $f \circ f \circ \dots \circ f$. **Complex dynamics** and **arithmetic dynamics** are branches of mathematics that study this.
- Consider the function

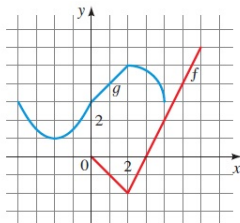
$$f : \mathbb{N}^* \longrightarrow \mathbb{N}^*$$

$$n \longmapsto \begin{cases} \frac{n}{2} & \text{if } n \text{ is even,} \\ \frac{3n+1}{2} & \text{if } n \text{ is odd.} \end{cases}$$

Start with any non-zero natural number n you like (choose a really big one to make it interesting) and compute $f(n)$, $(f \circ f)(n)$, $(f \circ f \circ f)(n)$, $(f \circ f \circ f \circ f)(n)$, $\dots$, you will get 1 eventually. This is known as **the Collatz conjecture** and it hasn't been proven ever since it was introduced in 1937.

ACTIVITY 3

- (a) Given $f(x) = x^2 - 1$, $g(x) = \frac{1}{|x|}$, and $h(x) = \sqrt{x-3}$, find the following composite functions with their domains.
- (i) $f \circ g$
- (ii) $g \circ f$
- (iii) $f \circ h$
- (iv) $f \circ g \circ h$
- (b) Use the given graphs of f and g to evaluate the following expressions.



- $$\begin{array}{ll} \text{(i)} & (f \circ g)(2) \\ \text{(ii)} & (g \circ f)(0) \end{array} \qquad \begin{array}{ll} \text{(iii)} & (g \circ g)(-2) \\ \text{(iv)} & (g \circ f \circ f)(4) \end{array}$$

ACTIVITY 3: SOLUTION

(a) $f(x) = x^2 - 1$, $g(x) = \frac{1}{|x|}$, $h(x) = \sqrt{x-3}$. The domains of f, g, h are $\mathbb{R}, \mathbb{R} \setminus \{0\}$ and $[3, \infty)$ respectively.

$$(i) (f \circ g)(x) = f(g(x)) = f\left(\frac{1}{|x|}\right) = \left(\frac{1}{|x|}\right)^2 - 1 = \frac{1}{x^2} - 1.$$

Since for all $x \in \mathbb{R} \setminus \{0\}$, $g(x) \in \mathbb{R} = \text{domain of } f$, the domain of $f \circ g$ is $\mathbb{R} \setminus \{0\}$.

$$(ii) (g \circ f)(x) = g(f(x)) = g(x^2 - 1) = \frac{1}{|x^2 - 1|}. \text{ For } f(x) \in \mathbb{R} \setminus \{0\} = \text{domain of } g, \text{ we need } f(x) \neq 0, \text{ i.e. } x \neq \pm 1. \text{ Hence, the domain of } g \circ f \text{ is } \mathbb{R} \setminus \{-1, 1\}.$$

$$(iii) (f \circ h)(x) = f(h(x)) = f(\sqrt{x-3}) = (\sqrt{x-3})^2 - 1 = x - 3 - 1 = x - 4. \text{ Since for all } x \in [3, \infty), h(x) \in \mathbb{R} = \text{domain of } f, \text{ the domain of } f \circ h \text{ is } [3, \infty).$$

ACTIVITY 3: SOLUTION

(a) $f(x) = x^2 - 1$, $g(x) = \frac{1}{|x|}$, $h(x) = \sqrt{x-3}$. The domains of f, g, h are $\mathbb{R}, \mathbb{R} \setminus \{0\}$ and $[3, \infty)$ respectively.

$$(iv) (f \circ g \circ h) = f(g(h(x))) = f(g(\sqrt{x-3})) =$$

$$f\left(\frac{1}{|\sqrt{x-3}|}\right) = \left(\frac{1}{|\sqrt{x-3}|}\right)^2 - 1 = \frac{1}{x-3} - 1. \text{ For}$$

$h(x) \in \mathbb{R} \setminus \{0\} = \text{domain of } g$, we need $h(x) \neq 0$, i.e.

$x \in [3, \infty) \setminus \{3\} = (3, \infty)$, which is the domain of $g \circ h$.

Since for all $x \in (3, \infty)$, $(g \circ h)(x) \in \mathbb{R} = \text{domain of } f$,
the domain of $f \circ g \circ h$ is $(3, \infty)$.

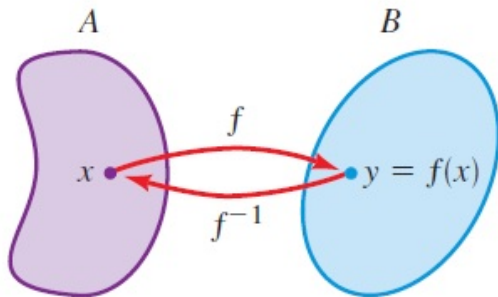
⚠ As illustrated in the last two problems, we cannot deduce the domain of a composite function solely based on its expression.

OUTLINE

- 1 Transformations of Functions
- 2 Combining Functions
- 3 One-to-One, Onto and Inverses
- 4 Summary

MOTIVATION OF INVERSE

Intuitively, the inverse f^{-1} of a function $f : A \longrightarrow B$ is the function that reverses the process or undoes the effect of f :



E.g.: For $f(x) = x + 5$, the inverse $f^{-1}(x) = x - 5$, because to undo the effect of $+5$, we need to -5 .

THE INVERSE OF A FUNCTION

DEFINITION:

Let $f : X \longrightarrow Y$ be a function. If $g : Y \longrightarrow X$ is another function such that

for all $x \in X$, $g(f(x)) = x$ and for all $y \in Y$, $f(g(y)) = y$,

then $g(x)$ is called the **inverse function** of $f(x)$ and it's denoted as $f^{-1}(x)$ (do not confuse this notation with $(f(x))^{-1} = \frac{1}{f(x)}$). In this case, f is said to be **invertible**.

However, not all functions have an inverse. For example $f : \mathbb{R} \longrightarrow \mathbb{R}$, $f(x) = x^2$ doesn't have an inverse. This is because when we reverse the process of f , we might get more than one value. For instance, f sends both 2 and -2 to 4, so should the inverse of f send 4 to 2 or -2? We can't choose both as the inverse is a function and it must return only one value for each input. Also, we can't reverse the process of f on negative $y \in \mathbb{R}$.

ONE-TO-ONE AND ONTO FUNCTIONS

The reason that $f(x) = x^2$ can't have an inverse is because it maps two (or more) different x 's to the same value and it doesn't map onto its entire codomain. If a function never maps two or more different x 's to the same value and its range equals to its codomain, then it has an inverse. This motivates us to define the following:

DEFINITION:

A function $f : X \longrightarrow Y$ is **one-to-one**/**injective** if for all $x_1, x_2 \in X$,

$$x_1 \neq x_2 \quad \Rightarrow \quad f(x_1) \neq f(x_2),$$

i.e. f never maps two or more different x 's to the same value.

DEFINITION:

A function $f : X \longrightarrow Y$ is **onto**/**surjective** if its range equals to its codomain.

THEOREM OF INVERTIBLE FUNCTIONS

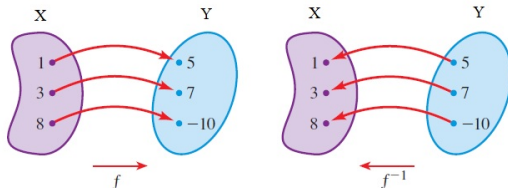
THEOREM OF INVERTIBLE FUNCTIONS

A function $f : X \longrightarrow Y$ is one-to-one and onto iff. f has an inverse. Explicitly,

$$f^{-1} : Y \longrightarrow X$$

$y \longmapsto$ the unique $x \in X$ such that $f(x) = y$.

E.g.: $f : X = \{1, 3, 8\} \rightarrow Y = \{5, 7, -10\}$, where $f(1) = 5$, $f(3) = 7$ and $f(8) = -10$. So f is one-to-one and onto, thus it has an inverse.



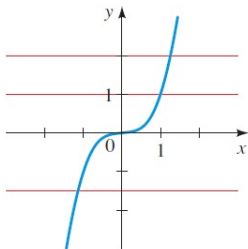
HORIZONTAL LINE TEST

The one-to-one property can be geometrically translated into the horizontal line test:

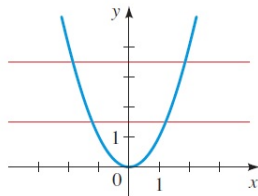
HORIZONTAL LINE TEST

A function is one-to-one if and only if no horizontal line intersects its graph more than once.

E.g.:



$f(x) = x^3$ is one-to-one.



$g(x) = x^2$ is not one-to-one.

ACTIVITY 4

Determine whether the function

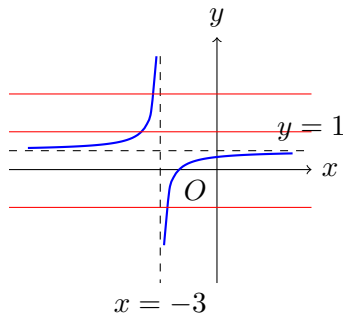
$$f : \mathbb{R} \setminus \{-3\} \longrightarrow \mathbb{R} \setminus \{1\}$$

$$x \longmapsto \frac{x+2}{x+3}$$

from activity 1 is one-to-one and onto, and hence conclude whether it's invertible.

ACTIVITY 4: SOLUTION

By applying the horizontal line test to the graph of $y = f(x)$,



we see that $f(x)$ is one-to-one. From the graph also, we see that the range of f is $\mathbb{R} \setminus \{1\} = \text{codomain of } f$, so f is onto. Thus, by the theorem of invertible functions, f is invertible.

FINDING THE INVERSE

To find the rule or the expression of an inverse function, we can try the following approach:

1. Write $y = f(x)$.
2. Solve this equation by writing x in terms of y (if possible).
3. The expression in terms of y from the previous step is $f^{-1}(y)$.
Since we normally write a function (inverse functions are functions) in terms of x , so the rule of the inverse function is $x \mapsto f^{-1}(x)$, i.e. we replace y by x .

E.g.: We can check easily that $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = 2x - 1$ is invertible (it's one-to-one and onto). So

$$y = 2x - 1 \Leftrightarrow x = \frac{y + 1}{2} = f^{-1}(y).$$

Thus, we have $f^{-1} : \mathbb{R} \rightarrow \mathbb{R}$, $f^{-1}(x) = \frac{x+1}{2}$.

ACTIVITY 5

Find the rule of the inverse of

$$f : \mathbb{R} \setminus \{-3\} \longrightarrow \mathbb{R} \setminus \{1\}$$

$$x \longmapsto \frac{x+2}{x+3}.$$

ACTIVITY 5: SOLUTION

We have proven f is invertible from the previous activity. To find the rule of the inverse, we try to express x in terms of y :

$$y = \frac{x+2}{x+3} = \frac{(x+3)-1}{x+3} = 1 - \frac{1}{x+3}$$

$$\Leftrightarrow \frac{1}{x+3} = 1 - y$$

$$\Leftrightarrow x+3 = \frac{1}{1-y}$$

$$\Leftrightarrow x = \frac{1}{1-y} - 3 = \frac{1-3(1-y)}{1-y} = \frac{3y-2}{1-y}.$$

Thus, we have

$$\begin{aligned} f^{-1} : \mathbb{R} \setminus \{1\} &\longrightarrow \mathbb{R} \setminus \{-3\} \\ x &\longmapsto \frac{3x-2}{1-x}. \end{aligned}$$

FINDING THE INVERSE

Fun fact: It's possible that we can't express x in terms of y algebraically even though we can prove that the function is invertible. For example, $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x + \sin(x)$ is such a function. Once you have learned differential calculus, you can show that $f(x) = x + \sin(x)$ is increasing, hence one-to-one. On the other hand, by considering the limits when $x \rightarrow \pm\infty$, you can conclude that f is onto. So f is indeed invertible. But you can't express x in terms of y algebraically.

ACTIVITY 6

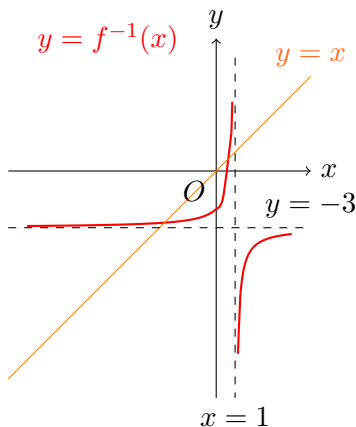
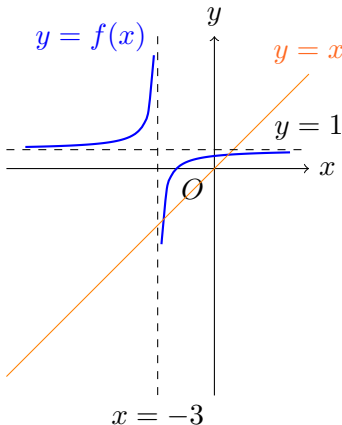
Sketch the graph of the inverse of

$$f : \mathbb{R} \setminus \{-3\} \longrightarrow \mathbb{R} \setminus \{1\}$$

$$x \longmapsto \frac{x+2}{x+3}.$$

ACTIVITY 6: SOLUTION

Note that the asymptotes $x = -3$ is reflected to $y = -3$ and the asymptotes $y = 1$ is reflected to $x = 1$.



OUTLINE

- ① Transformations of Functions
- ② Combining Functions
- ③ One-to-One, Onto and Inverses
- ④ Summary

SUMMARY

- Transformations of Functions
 - vertical and horizontal shifting, reflecting, vertical and horizontal stretching and shrinking.
 - absolute value of a function, absolute value of the input.
 - odd and even functions.
- Combining Functions
 - sums, differences, products, and quotients.
 - composite functions.
- One-to-One, Onto and Inverses
 - the inverse of a function.
 - one-to-one and onto functions.
 - Theorem of invertible functions.
 - horizontal line test.
 - finding the inverse.
 - the graph of the inverse of a function.

RESOURCES

References:

- James Stewart. Precalculus Mathematics for Calculus (7th edition) sections 2.6, 2.7, 2.8.